

Pathology Foundation Models are Scanner Sensitive: Benchmark and Mitigation with Contrastive *ScanGen* Loss

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Abstract. Computational pathology (CPath) has shown great potential in mining actionable insights from Whole Slide Images (WSIs). Deep Learning (DL) has been at the center of modern CPath, and while it delivers unprecedented performance, it is also known that DL may be affected by irrelevant details, such as those introduced during scanning by different commercially available scanners. This may lead to *scanner bias*, where the model outputs for the same tissue acquired by different scanners may vary. In turn, it hinders the trust of clinicians in CPath-based tools and their deployment in real-world clinical practices. Recent pathology Foundation Models (FMs) promise to provide better domain generalization capabilities. In this paper, we benchmark FMs using a multi-scanner dataset and show that FMs still suffer from scanner bias. Following this observation, we propose *ScanGen*, a contrastive loss function applied during task-specific fine-tuning that mitigates scanner bias, thereby enhancing the models’ robustness to scanner variations. Our approach is applied to the Multiple Instance Learning task of Epidermal Growth Factor Receptor (EGFR) mutation prediction from H&E-stained WSIs in lung cancer. We observe that *ScanGen* notably enhances the ability to generalize across scanners, while retaining or improving the performance of EGFR mutation prediction.

Keywords: Computational Pathology · Scanner Generalization · Foundation Models.

1 Introduction

Digital pathology is catalyzing a new era in the field of histopathology by transforming how clinicians interact with and analyze tissue samples. Furthermore, it enables computational pathology (CPath) over whole slide images (WSIs), a key component for deriving novel and quantitative insights from WSIs. This brings

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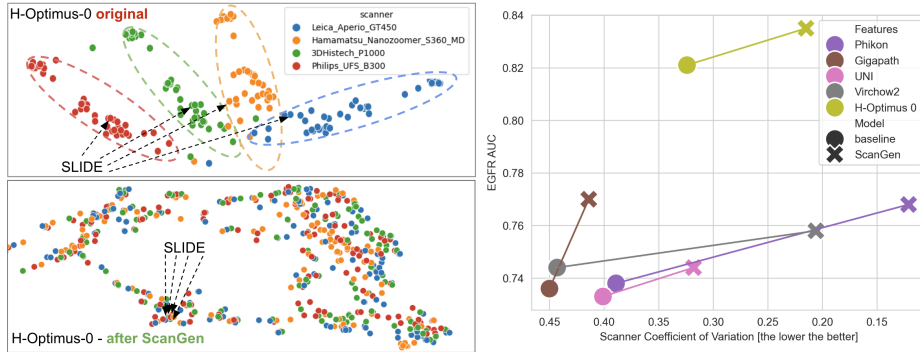


Fig. 1. Left: Existing pathology Foundation Models suffer from scanner bias. We show a 2D projection of WSIs (dots) from 4 scanners (colors) using UMAP based on H-Optimus-0 embeddings before and after *ScanGen*. "SLIDE→" indicates the slides of the same specimen from different scanners. Right: *ScanGen* improves both generalization and performance.

new capabilities to clinical practice, such as prediction of genetic alterations [11, 22], tumor micro-environment analysis [16], or survival prediction [18].

Digital pathology requires the digitization of tissue slides into WSIs using slide scanners. While these scanners must capture the underlying tissue accurately, different devices may result in differences in image characteristics [5]. For example, differences in optical systems may impact image sharpness, variations in light sources can affect contrast and introduce artifacts, and proprietary software (e.g. for data compression) can affect image processing. To implement CPath-based tools in clinical settings, it is crucial to address the constraints specific to different scanners. Since different scanners are used worldwide, with some being predominant in certain regions (e.g., Hamamatsu in Asia [17]), any performance issues associated with a particular scanner could lead to regional disparities in patient care.

Recently, the introduction of foundation models (FMs) provides a base upon which CPath tools are developed. However, while FMs promise better generalization capabilities, they may still encode several biases, such as the slide identity [24] and hospital environments [9]. The present paper delves into another crucial aspect of CPath, the digital scanners used during image acquisition, offering empirical evidence that FMs are sensitive to the type of digital scanner employed for reading tissue specimens. Since CPath FMs are typically used as frozen extractors with no access to the input images, alleviating this bias with traditional stain normalization methods (e.g., Reinhard[19], Macenko [14]) or DL ones (e.g., StainGAN[21], StainNet[10]) is not trivial. Scanner bias results in weakened generalization capabilities for FM-based CPath models, posing challenges in their broader application. In particular, scanner bias means that the model output will change based on which scanner has been used to read the specimen despite

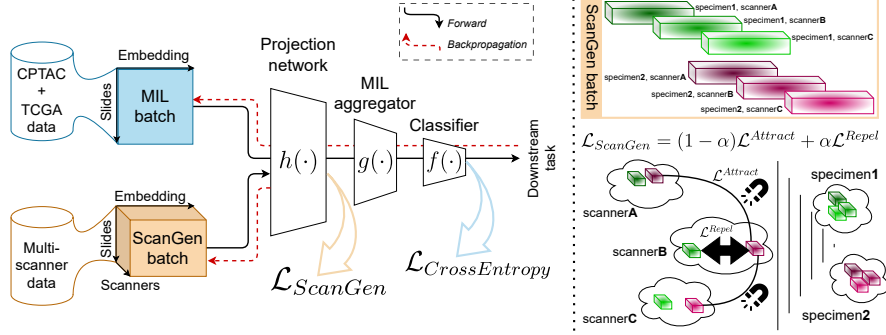


Fig. 2. Our method. Left: Data are projected to a new embedding space where scanner bias is removed via our contrastive *ScanGen* loss. *ScanGen* is trained on paired WSIs of the same specimen acquired with different scanners and can be easily integrated with any existing method by acting before the MIL aggregator. Right: *ScanGen* attracts the representations of same-specimen different-scanner pairs, while repelling same-scanner different-specimen ones. This favors specimen-based clusters to scanner-based ones, thus fostering generalization across scanners in downstream applications.

the content being the same, which can significantly hinder trust in CPath-based tools and, consequently, slower clinical adoption.

To address this shortcoming, we propose *ScanGen*, a contrastive loss function that mitigates scanner bias, thereby enhancing the model’s robustness across different scanners. Our evaluation involves testing various state-of-the-art FMs using a benchmark set that comprises images of the same specimen captured from six commercially available scanners at $40\times$ magnification and one scanner at $20\times$ magnification. Additionally, our approach is applied to the task of detecting Epidermal Growth Factor Receptor (EGFR) mutations in lung cancer, a prevalent and clinically significant endeavor in the CPath space [2, 25].

In summary, this paper introduces three contributions: 1) a contrastive loss function aimed at reducing scanner bias in Multiple Instance Learning (MIL), 2) an assessment of FMs in relation to scanner generalization, and 3) application of the proposed method to EGFR mutation prediction in lung cancer, a clinically relevant task. The results demonstrate promising improvements in predictive performance and generalization (Fig. 1), underscoring the potential of our methodologies in clinical settings.

2 Method

2.1 Scanner Generalization Metric for MIL

Due to the extremely large size of WSIs, MIL is usually employed for classification or regression tasks. In MIL, WSIs are partitioned into P patches and the $FM(\cdot)$ is used as a pretrained, frozen feature extractor to obtain an embedding

vector $x_i = FM(p_i)$ per each patch p_i . Finally, an aggregator function $g(\cdot)$ combines all patches and a classifier $f(\cdot)$ produces the final prediction (e.g., EGFR mutation) as $\hat{y} = f(g(x_1, \dots, x_P))$. In this context, the scanner bias relates to how different two predictions $\hat{y}_m^{s_1}$ and $\hat{y}_m^{s_2}$ are from the same specimen m but different scanners s_1 and s_2 . For this reason, we measure the intensity of scanner bias as the average coefficient of variation (CoV) between predictions of the same specimen m across S multiple scanners, defined as

$$\text{CoV} = \frac{1}{M} \sum_{m=1}^M \left(\frac{\sigma(\hat{y}_m^{s_1}, \hat{y}_m^{s_2}, \dots, \hat{y}_m^{s_S})}{|\mu(\hat{y}_m^{s_1}, \hat{y}_m^{s_2}, \dots, \hat{y}_m^{s_S})|} \right), \quad (1)$$

with M number of specimens, where σ and μ are, respectively, the standard deviation and average across the logit predictions of the same specimen acquired with different scanner hardware. CoV is more appropriate than simple standard deviation because it allows for meaningful comparisons across sources with different scales. Lower CoV means better agreement in predictions, indicating the model is less scanner-sensitive.

2.2 Proposed Framework and *ScanGen* Loss

As Figure 2 illustrates, to reduce the scanner bias in the embedding space, we first map the patch embeddings x to a new scanner-unbiased embedding space using a projection network $h(\cdot)$, which is trained as follows. Given two specimens m and n and two scanners s_1 and s_2 , we rework the contrastive loss function[7] and define our *ScanGen* loss as

$$\mathcal{L}_{\text{ScanGen}} = (1 - \alpha) \underbrace{(d(h(x_m^{s_1}), h(x_m^{s_2})))^2}_{\mathcal{L}^{\text{Attract}}} + \alpha \underbrace{(\max(0, r - d(h(x_m^{s_1}), h(x_n^{s_1}))))^2}_{\mathcal{L}^{\text{Repel}}} \quad (2)$$

where $d(a, b) = 1 - \frac{a \cdot b}{|a| \cdot |b|}$ is the cosine distance function, $\mathcal{L}^{\text{Attract}}$ is the attraction term encouraging samples from the same specimen but different scanners to be close in the embedding space, $\mathcal{L}^{\text{Repel}}$ is the repulsion term pushing samples from the same scanner but different specimens apart, r is the radius, and α is a hyper-parameter governing the weighted combination of those terms. The MIL aggregator is then applied: $g(\cdot) = g(h(x_1), h(x_2), \dots, h(x_P))$.

The final loss is composed of the downstream task loss, in our case EGFR prediction using standard cross-entropy (CE), and our *ScanGen* loss:

$$\mathcal{L} = \frac{1}{|A|} \sum_{a \in A} \mathcal{L}_{\text{CE}}(\hat{y}_a, y_a) + \lambda \cdot \frac{1}{|B|} \sum_{\substack{m, n \in B \\ m \neq n}} \sum_{\substack{u, v \in S \\ u \neq v}} \mathcal{L}_{\text{ScanGen}}(x_m^{s_u}, x_m^{s_v}, x_n^{s_u}; h) \quad (3)$$

where A and B are independent sets of labeled samples for the downstream task and the *ScanGen* task, respectively, S is the number of scanners, and λ is a hyper-parameter to control the balance between those two terms.

2.3 Implementation

Within a shared model, two different data flows occur at each forward pass during training (see Fig. 2-left). On the one hand, a regular MIL batch sampled from an EGFR-labeled data pool enters the projection-aggregation-classification cascade, thus producing the logits for standard CE loss computation.

On the other hand, a batch of embeddings identified by specimen and scanner names is fed into the projection network $h(\cdot)$. There, the *ScanGen* loss first calculates a distance matrix using pairwise cosine similarity between embeddings and constructs matrices to identify, within the batch, pairs of samples from the same scanner or with the same specimen name. These matrices are used to compute the attraction and repulsion losses, which are weighted and combined on the basis of the α parameter. The final loss is averaged over valid pairs, ensuring that only relevant pairs (i.e., either same-specimen different-scanners or same-scanner different-specimen) contribute to the loss calculation. For fair comparison, also the baseline models where *ScanGen* is not enabled utilize the projection network $h(\cdot)$, which is learned, together with $g(\cdot)$ and $f(\cdot)$, under the CE loss alone.

We developed our code in PyTorch. We implement $h(\cdot)$ as a multi-layer perceptron (MLP) with three layers and hidden dimension between 48 and 96 depending on the FM features. Regarding $g(\cdot)$, we use the global average pooling (GAP) as a simple MIL model, where patch-level features are averaged to produce one embedding per WSI: $X = \frac{1}{P} \sum_{i=1}^P x_i$, followed by an MLP classifier. Moreover, Tab. 2 shows an adaptation to some modern MIL aggregators, for which we followed their official implementation. In our experiments, we found that α of 0.12 to 0.21, r of 0.90 to 1.10, and λ of 0.5 to 1.0 are appropriate.

3 Results

3.1 Datasets

Two publicly available datasets, CPTAC [4] and TCGA [1]), for a total of 1,144 WSIs, were used to train the EGFR branch. Note that only non-small cell lung cancer specimens were selected. Our multi-scanner dataset, on the other hand, is composed of 323 specimens, each tested for EGFR mutation using PCR and NGS assay as well as scanned using six devices: Leica Aperio AT2, Leica Aperio GT450, Hamamatsu Nanozoomer S360MD, 3DHistech P1000, Philips UFS B300, Roche Ventana DP200. The specimens were scanned at $40\times$ magnification. The specimens were also scanned with Leica Aperio GT450 at $20\times$ magnification. In total, the multi-scanner dataset is composed of $323 * 7 = 2,261$ WSIs. Fig. 4-right shows different scans of the same specimen using the six different scanners. We split this dataset into three splits while preserving class balance and specimen-level split (i.e., WSIs of the same specimen were grouped). As a result, 54 specimens were used to train the *ScanGen* branch, 45 specimens were used as validation set for tuning both EGFR branch and *ScanGen* branch, and 224 specimens were used as held-out test set for external evaluation of downstream task performance (EGFR AUC) and scanner agreement (CoV).

Table 1. Evaluating scanner generalization for available foundation models on held-out testset. Coefficient of EGFR Variation (CoV) is shown across six scanners and two magnifications (40 $\times$, 20 $\times$).

Embedding	Use <i>ScanGen</i>	EGFR AUC $\uparrow$	Scanner CoV $\downarrow$	Magnification CoV $\downarrow$
Phikon[6]	No	0.738	0.389	0.158
	Yes	0.768	0.121 (-68.9%)	0.053 (-66.4%)
UNI[3]	No	0.733	0.401	0.116
	Yes	0.744	0.318 (-20.7%)	0.089 (-23.3%)
Virchow[26]	No	0.744	0.443	0.125
	Yes	0.758	0.206 (-53.5%)	0.077 (-38.4%)
Gigapath[23]	No	0.736	0.450	0.174
	Yes	0.770	0.414 (-8.00%)	0.193 (+10.9%)
H-Optimus-0[20]	No	0.821	0.324	0.117
	Yes	0.835	0.215 (-33.6%)	0.108 (-7.70%)

3.2 Evaluating Scanner Generalization

We assess FMs on our multi-scanner dataset, presenting both qualitative and quantitative outcomes. To analyze the bias for each FM visually, for each WSI in the test set, we average all patch embeddings to produce a single WSI embedding, and project the full test set in 2D using UMAP [15] with cosine metric and default parameters. Fig. 3 illustrates that each FM is affected by scanner bias, as evidenced by the clustering and separation of samples according to the scanner rather than the same specimen. The only exception is UNI [3], which, according to the 2D projection, seems to be unaffected by this bias; however, our quantitative results paint a different picture, and even UNI largely benefits from our *ScanGen* loss.

In Tab. 1, we benchmark five state-of-the-art FMs according to our new metrics for evaluating generalization, CoV over scanners and CoV over magnifications, and show *ScanGen*’s mitigation effect. Our expectation is that *ScanGen* would reduce the CoV, while avoiding hindrance on the EGFR downstream task. Our findings indicate that incorporating *ScanGen* projection significantly enhances the ability to generalize across scanners (lower CoV). More importantly, the predictive performance for EGFR is not hindered; rather, it actually improves (higher AUC) for each embedding type. Additionally, according to our findings, the *ScanGen* loss also helps mitigate a disagreement in predictions when different magnifications are used to read the slide.

A similar result is shown in Fig. 1-right, where each model is compared on CoV and EGFR AUC, with and without *ScanGen* loss. While CoV significantly improves for each FM, we also see a boost in EGFR AUC.

3.3 *ScanGen* with Limited Scanners

This section presents an ablation study evaluating the impact of varying the number of scanner types used for *ScanGen* loss during model training. Figure 4

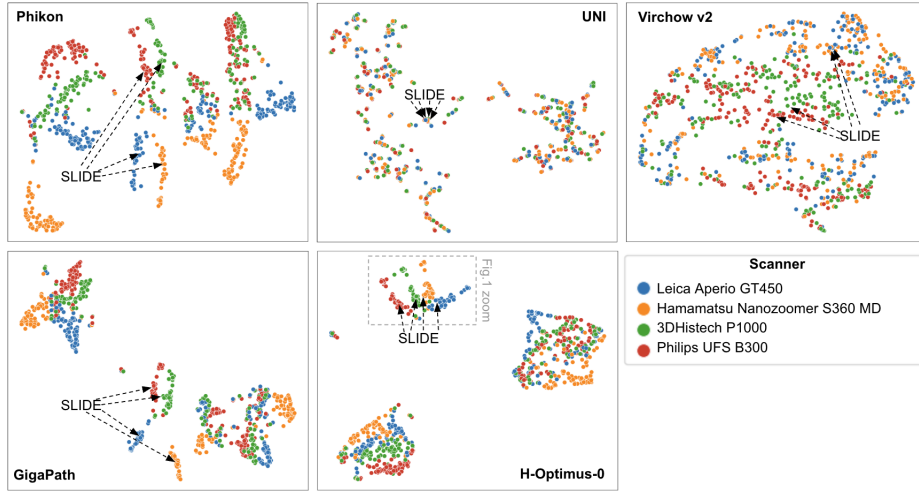


Fig. 3. Our multi-scanner dataset projected to 2D using various FMs. Each dot is a WSI embedding from the test set by averaging FM embeddings across patches and projecting them using UMAP. The color indicates the scanner. We selected 4 out of 7 scanners to reduce clutter. A single slide is indicated with "SLIDE→" to emphasize the location of the slides of the same specimen from different scanners in each plot.

highlights the gain of CoV compared to the baseline. Each model is tested on the same external test set containing all six available scanner sources. It examined scenarios where different combinations of two to six scanner types were utilized for the training process. The plot indicates the monotonic correlation between the number of scanner types and the scanner generalization performance: when more scanner types are used for the *ScanGen* loss, the generalization across scanners increases. Performance improvement converges at five scanners. Nevertheless, even when only three scanner types are used for training, the scanner generalization on six test scanners significantly improves by 27%.

This ablation study also allowed us to investigate which scanner combinations led to the highest or lowest CoV improvement over baseline. We counted how often a certain scanner was present in the best (i.e., lowest CoV) and worst combinations. Doing this for all ablation scenarios from two to five scanners, we found that Leica Aperio GT450 and the Hamamatsu scanners were *always* in the best performing configurations, while the Leica Aperio AT2 and GT450 pair was the one most often associated with the worst performing configurations. This suggests that using a diverse set of scanners for *ScanGen* loss leads to the best generalization, while using similar scanners, like the two Leica Aperio, does not benefit scanner robustness much.

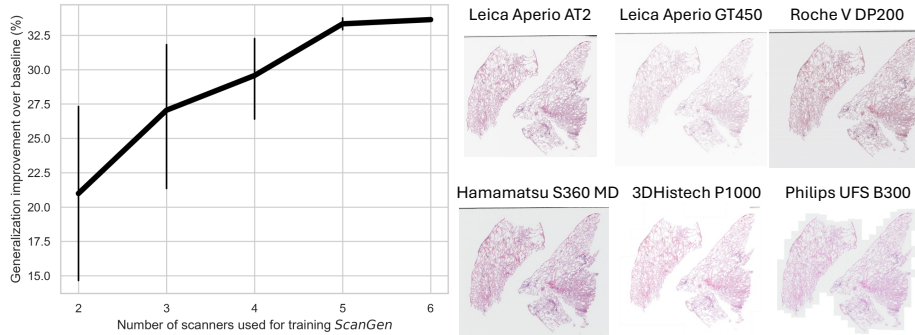


Fig. 4. Left: Scanner agreement improves with the number of scanners used for training *ScanGen*. Vertical bars indicate the standard deviation across scanner combinations. Right: Examples of the same specimen digitized using different scanners.

3.4 Extension to Other MIL Methods

We also showcase our method’s performance on recent MIL techniques like AB-MIL [8], DS-MIL [13], and SlotMIL [12]. Table 2 summarizes our findings with H-Optimus-0 features. Our *ScanGen* loss consistently reduced CoV, boosting agreement by 11% to 33%, and maintained or surpassed the baseline AUC.

Table 2. Integrating our *ScanGen* loss into other MIL approaches. We report the AUC and scanner CoV values obtained on the held-out test set.

MIL approach	GAP		AB-MIL[8]		DS-MIL[13]		SlotMIL[12]	
	AUC	CoV [↓]	AUC	CoV [↓]	AUC	CoV [↓]	AUC	CoV [↓]
EGFR	0.821	0.324	0.828	0.251	0.834	0.265	0.822	0.338
EGFR + <i>ScanGen</i>	0.835	0.215	0.835	0.224	0.828	0.214	0.826	0.227

4 Discussion

This study demonstrates that current FMs for CPath are sensitive to scanners, impacting the model results. In the global market, scanner distribution is uneven. Leica and Philips lead in America, 3DHistech challenges Leica in Europe, and Hamamatsu is dominant in Asia [17]. Differences in device performance may affect quality or care between Asia, the U.S., and Europe. To build real-world CPath tools on a global scale, we need to think about generalization across hospitals in different geographical regions, which requires more robust predictive models.

Our research is essential in emphasizing a current problem stemming from the swift adoption of FMs in numerous CPath applications. We first gathered a

benchmark comprising six widely used scanners available in the market. Then, we presented empirical evidence of scanner bias across various cutting-edge FMs. Finally, we offered a solution to mitigate the bias in existing FMs by integrating a scanner generalization loss on top of the standard MIL downstream task. Ultimately, we aspire that our efforts will motivate future FM developers to address this problem during model training. Nonetheless, training an FM is beyond the reach of most research labs due to the necessity of tens of thousands of samples and significant computational effort. Consequently, our initial focus is on demonstrating the presence of the issue and suggesting a cost-effective solution to enhance the robustness of existing open-source FMs against scanner bias.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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